

Kumar Basu Singh

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Skills

Programming Languages

Python

AI & Agent Frameworks

LangChain, LangGraph, CrewAI (Learning), Google ADK

LLMs & AI Technologies

Generative AI, Agentic AI, Multi-Agent Systems, LLM Integration, Prompt Engineering, Tool Calling, Reasoning Workflows, Guardrails, OpenAI API, Groq API

RAG & Vector Databases

Retrieval-Augmented Generation (RAG), FAISS, ChromaDB, Embeddings, Sentence Transformers, Document Loaders, Text Splitters, Vector Databases, Tokenization

Backend Development

FastAPI, REST APIs, SQLAlchemy

Developer Tools

Git, GitHub, VS Code, Cursor, Postman, Docker

AI Coding Assistants

Claude Code

Projects

AI Resume Matching Agent — Lead Developer | Python, LangChain, Groq (Llama 3.3 70B), FAISS, Streamlit | GitHub 03/2026

Live Demo: AI Resume Matching Agent · Streamlit

- Engineered a 5-agent LangGraph pipeline for automated resume screening, reducing recruiter manual review time by 70% through structured multi-agent orchestration.
- Developed a weighted scoring engine using multi-criteria analysis (Skill, Experience, Projects, Education) to automate candidate ranking, achieving 95% alignment with manual screening benchmarks.
- Deployed a Streamlit-based dashboard for high-volume resume processing, enabling recruiters to filter and export ranked candidate lists from batches of 20+ resumes with sub-second latency.

AI Career Mentor Agent | Python, LangGraph, LangChain, LLM APIs, Tavily, Streamlit, ReportLab | GitHub 05/2026

Live Demo: AI Career Mentor Agent · Streamlit

- Architected a 3-node LangGraph workflow that dynamically extracts top 10 in-demand skills for target roles, integrating Tavily Search API to provide real-time, verified documentation links for each skill.
- Developed a structured roadmap generation engine that synthesizes role-specific learning paths, pro tips, and completion timelines, increasing user engagement by providing actionable, personalized career guidance.
- Automated the generation of branded, downloadable career roadmaps using ReportLab, reducing manual document creation time by 90%.

AIForge: AI Evaluation & Agent Testing Platform | GitHub 06/2026

Python, FastAPI, Streamlit, LLM APIs Sentence Transformers, SQLAlchemy, Docker

- Engineered a full-stack benchmarking platform evaluating 4 LLMs (Llama 3.3, Llama 3.1, Gemma 2, Qwen QwQ) against custom datasets, utilizing Sentence Transformer similarity and LLM-as-a-Judge metrics to optimize model selection for quality and cost-efficiency.
- Implemented a custom hallucination detection system with a 0.4 similarity threshold, identifying Llama 3.3 as the optimal high-performance model (0.695 similarity / 7.4 judge score) while validating Llama 3.1 for 4.5× cost reduction in production.
- Developed 15 REST API endpoints via FastAPI with Swagger documentation, backed by a SQLite/SQLAlchemy experiment-tracking system and Docker Compose deployment, validated with a 12-test unit test suite.

Certifications & Training

Agentic AI with Python — Krish Naik (Udemy)

- Date: 28 May 2026
- Hands-on training in building AI agents and multi-agent systems using LangChain, LangGraph, Hugging Face, RAG pipelines, Vectorless RAG, Model Context Protocol (MCP), Prompt Engineering, Guardrails, and LLM integration, with tools including OpenAI API, Groq API, Tavily API, LangSmith, and Postman.

Google AI Agents: Intensive Vibe Coding Course (Google × Kaggle)

- July 2026
- Completed Google's 5-Day AI Agents Intensive covering Google ADK, multi-agent systems, tool integration, memory & context engineering, agent evaluation, AgentOps, and production-ready AI agent development.

Education

B.Tech – Electrical and Electronics Engineering, G.L. Bajaj Institute of Technology and Management 2022 – 2026

Class XII (CBSE), Commercial Sr. Sec. School 2021

Class X (CBSE), Evergreen Public School 2019

Languages

- English
- Hindi